PROJECT REPORT / PREDICTIVE MAINTENANCE

NASA C-MAPSS

Exploratory analysis, FD001 model training and official test evaluation

Analysis and model results through 20 October 2026

LOWEST TEST RMSE	LOWEST TEST MAE	LOWEST NASA SCORE
24.050	**16.357**	**54.346 mean**
HistGradientBoosting	Random Forest	Ridge

EXECUTIVE TAKEAWAY

Causal rolling-feature regression is the tested core for this task. The saved FD001 models were evaluated on all 100 official test engine endpoints. Model ranking depends on the metric: HistGradientBoosting led RMSE, Random Forest led MAE, and Ridge led the NASA asymmetric score.

TinyML/quantization, neural sequence models, and the autoencoder are not measured competitors in these results. Their suitability and the practical limits of the user-proposed streaming update idea are assessed separately in this report.

Scope: all four subsets explored; FD001 alone trained and tested.
This is a regression study; classification accuracy and F1 are not applicable.

1. Problem framing and data analysis

Task. Estimate each engine’s remaining useful life in cycles from multivariate telemetry. Each row has unit ID, cycle, three operating settings, and 21 sensor channels (26 numeric columns total). Training trajectories run to failure; test trajectories stop early and are paired with one RUL label per engine.

What the data contains, what was checked, and what the plots support.

Subset	Train engines	Test engines	Operating conditions	Fault modes
FD001	100	100	1	1
FD002	260	259	6	1
FD003	100	100	1	2
FD004	248	249	6	2

All four subsets were loaded and checked in the executed notebook. The quality gate reported 0 missing and 0 non-finite cells in the train/test telemetry. The notebook also checked schema/label counts, duplicate unit-cycle keys, cycle ordering, trajectory lengths, and train/test endpoint differences.

- Compared subset complexity, operating-setting regimes, and trajectory sizes; FD002/FD004 have multiple operating regimes, so operating-state shifts should not automatically be interpreted as degradation.

- Screened sensor variability and constant/near-constant behavior; examined FD001 sensor trends, distributions, setting effects, correlations, and engine-to-engine variation.

- Constructed RUL targets and contrasted complete run-to-failure training trajectories with right-censored test endpoints; checked how validation/test endpoint conditions differ.

- Produced diagnostic plots across FD001–FD004 and preparation examples for causal features, engine-contained sequence windows, and early-life autoencoder windows.

Decision from EDA: start with FD001, which has one operating condition and one fault mode, then expand after a stable baseline. Do not transfer the FD001 result to FD002–FD004 without separate training and evaluation.

EDA COVERAGE

2. Training and test protocol

Separating model fitting, validation, and the official held-out test.

- Deterministic engine-wise split (seed 42): 80 development engines / 16,538 cycle rows for fitting; 20 held-out validation engines / 4,093 rows for selection. No engine crosses the split.
- 215 causal features: current cycle/settings/sensor measurements, cycles seen, and past-only sensor deltas, rolling means, standard deviations, and slopes over 5- and 20-cycle windows. Unit ID is excluded. Windows never cross engines.
- Uncapped RUL labels for fitting. Validation reports cycle-level scores plus endpoint simulations at 50%, 70%, and 90% of each validation engine trajectory to represent right-censoring.

Candidate	Role
Mean reference	Constant baseline; no learned sensor relationship
Ridge regression	Linear baseline using causal engineered features
Random Forest	Nonlinear tree ensemble on causal features
HistGradientBoosting	Boosted tree regressor on causal features

All four saved models were loaded and used only for prediction on FD001 test telemetry. For each of 100 engines, inference used its final observed cycle; the 100 supplied RUL labels were aligned by sorted unit ID and used only to calculate final metrics. No fitting or tuning was performed on official test data.

The official test has now been used as a final holdout evaluation. Do not tune these same model configurations to these test scores; future model selection should use development-only validation or a fresh untouched test set.

FD001 development and features

Models actually fitted and saved

Official test procedure

3. Measured model results

Validation and official-test metrics; lower is better for all shown metrics.

Model	RMSE	MAE	Mean NASA score
Mean reference	67.223	56.212	9274.472
Ridge regression	34.288	26.150	271.678
Random Forest	32.363	21.506	127.086
HistGradientBoosting	30.653	20.779	86.088

Model	RMSE	MAE	NASA total	NASA mean
HistGradientBoosting	24.050	16.885	33604.7	336.047
Random Forest	24.142	16.357	44880.7	448.807
Ridge regression	27.949	23.105	5434.6	54.346
Mean reference	52.796	40.996	276878.5	2768.785

Interpretation: the learned regressors substantially beat the mean reference on RMSE and MAE. On the official test, HistGradientBoosting has the lowest RMSE (24.050), Random Forest the lowest MAE (16.357), and Ridge the lowest mean NASA score (54.346). There is no single winner independent of the operational metric.

The NASA score penalizes late/optimistic RUL predictions more heavily than early/conservative predictions. The official-test error counts show overprediction more often than underprediction (HGB 66/34, RF 75/25, Ridge 75/25, mean 71/29), which helps explain why the NASA ranking differs from RMSE/MAE.

Accuracy, precision, recall, and F1 are classification metrics and are not valid for these continuous RUL estimates. No classification threshold or failure/no-failure labels were evaluated. R-squared was not reported in the completed experiment.

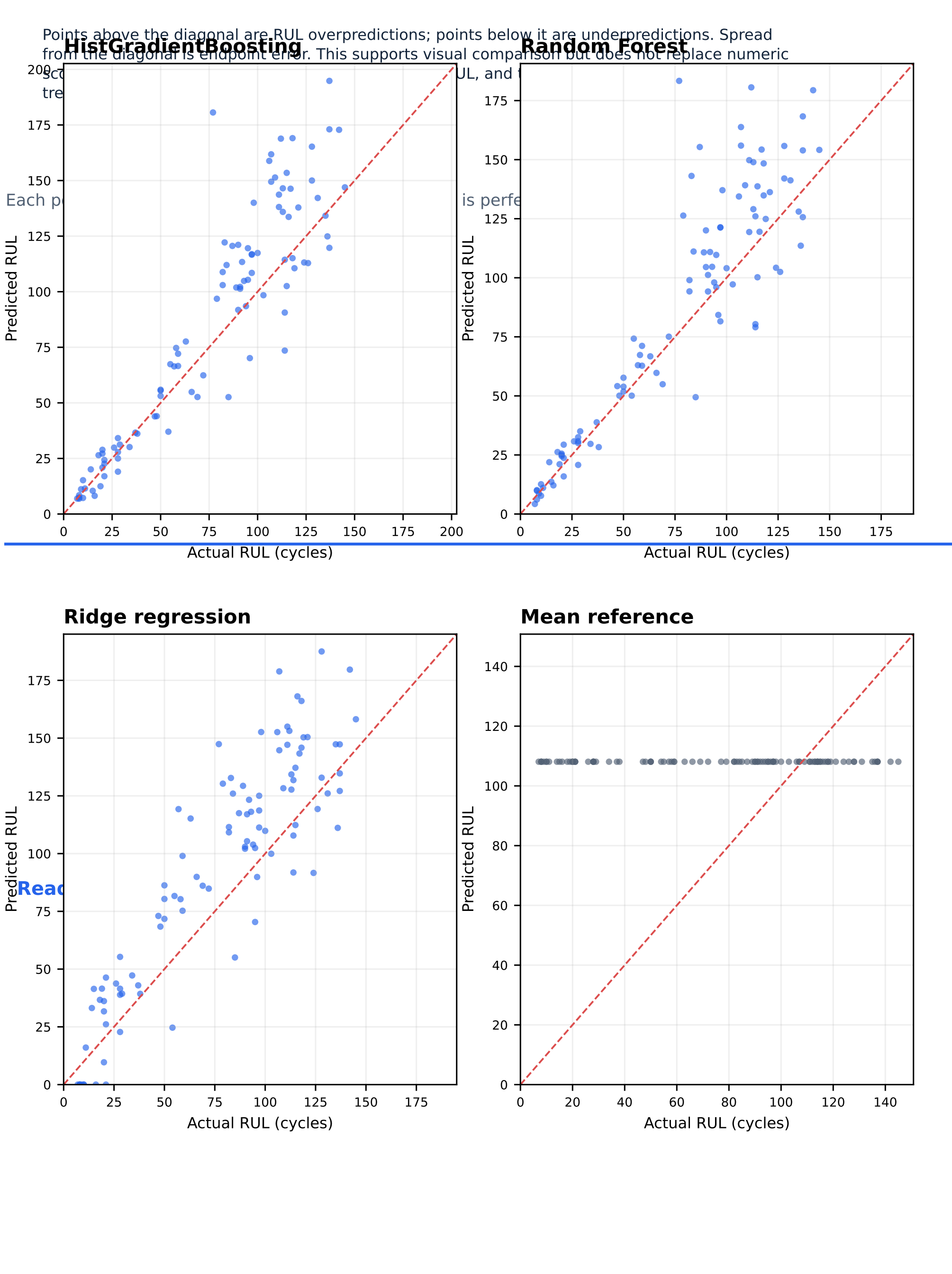
Held-out validation (4,093 cycle rows; 20 engines)

Official FD001 test (100 engine endpoints)

4. Official-test prediction diagnostics

Each p… is perfe…

Points above the diagonal are RUL overpredictions; points below it are underpredictions. Spread from the diagonal is endpoint error. This supports visual comparison but does not replace numeric sco… UL, and t… tre…



Reac…

5. Research methods and proposed TinyML/AE

What fits the problem, what was implemented, and what has actually been measured.

Approach	Assessment for this task	Status in this project
Research feature engineering + regression	Strong fit; causal rolling features and tree reg	215-feature causal frame used; AFICv selection itself not
AFICv (aggregated feature importance)	Useful candidate for reducing feature count a	Code path exists for development-only RF importance; n
TinyML / quantization	Possible deployment optimization after valida	No C-MAPSS TinyML model benchmark or quantization re
User-proposed neural + quantized AE stream	Potential edge inference/anomaly extension;	Not trained/evaluated on C-MAPSS; the earlier ~12 KB n
Autoencoder	Can learn a normal/early-life reconstruction b	Not trained in this experiment; no per-cycle anomaly lab
MLP / 1D CNN / LSTM	Reasonable future RUL comparisons if tempo	Optional code paths exist; these candidates remain untra

A sensor point alone does not reveal its true remaining life. Updating weights from the model's own prediction/error can reinforce mistakes. Online learning needs delayed trusted outcomes (for example, a verified failure-time label), safeguards against drift/catastrophic forgetting, and a controlled rollback path. Until then, edge inference plus local score/logging is safer than automatic self-training.

Use the research-backed feature-engineering + regression pipeline as the current C-MAPSS core. Treat autoencoder anomaly scoring and TinyML/quantization as separately evaluated extensions; do not claim they improve RUL until trained and compared on development-only validation.

Streaming-learning caveat

Overall judgment

6. Conclusions, limitations, and next steps

A decision-oriented summary of what the current evidence supports.

Conclusion

- The FD001 test results support causal sensor-history features with tree regressors as a credible first baseline. HGB and Random Forest are close in RMSE/MAE; Ridge’s NASA score advantage on this holdout reflects asymmetric penalties, not universally lower error.
- The test comparison covers 100 engines from one subset and only one deterministic development/validation split. There are no confidence intervals, repeated splits, statistical tests, FD002–FD004 trained models, or external fleet validation.
- The current code includes optional neural, sequence, autoencoder, AFICv, and TinyML utilities, but only mean, Ridge, Random Forest, and HistGradientBoosting were trained and tested with real C-MAPSS FD001 data.

Recommended next experiments

- Keep the official test untouched. Compare feature ablations, AFICv, capped-vs-uncapped targets, and hyperparameters using engine-grouped development validation only.
- If exploring the autoencoder, define a healthy/early-life reference using development engines and evaluate reconstruction-score behavior separately; obtain labeled anomalies before evaluating anomaly detection.
- If exploring TinyML, select a validated compact RUL regressor, quantize it, and measure latency, memory, and energy on the intended edge device. Report those separately from predictive scores.
- After FD001, train/evaluate separately on FD002–FD004, respecting multiple operating conditions and fault modes; do not assume the FD001 model generalizes.

Research and project sources

- NASA C-MAPSS dataset readme and case-study brief supplied with the project (26-column engine-cycle data; train-to-failure and right-censored test trajectories).
- Saxena, Goebel, Simon & Eklund (2008), “Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation,” PHM08.
- Ramasso & Saxena, “Review and Analysis of Algorithmic Approaches Developed for Prognostics on CMAPSS Dataset.” Used as broad RUL-method context.
- Alomari, Ando & Baptista, “Advancing aircraft engine RUL predictions: an interpretable integrated approach of feature engineering and aggregated feature importance.” Supports rolling features and AFICv as relevant candidate methods.

Related project artifacts: C-MAPSS_EDA_All_FD.ipynb; MODEL_TRAINING_DESIGN.md; MODEL_TRAINING_GUIDE.md; artifacts/FD001/official_test_metrics.csv; artifacts/FD001/official_test_predictions.csv.